Product Name: SANFUR - DS 125 (CEFUROXIME AXETIL POWDER FOR ORAL SUSPENSION USP 125mg/5ml)
Dosage form: Dry powder for oral Suspension.

Composition:
Each 5ml reconstituted suspension contains:
Cefuroxime Axetil USP equivalent to Anhydrous Cefuroxime 125mg.

Presentation:
Appearance of powder: White to off white coloured, granular powder with characteristic odour & flavour of Tutti Fruiti

Appearance after reconstitution: Off white coloured suspension.

Indications: SANFUR DS - 125 is an oral prodrug of the bactericidal cephalosporin antibiotic cefuroxime, which is resistant to most (beta)-lactamases and is active against a wide range of Gram-positive and Gram-negative organisms. It is indicated for the treatment of infections caused by susceptible bacteria. Susceptibility to SANFUR DS - 125 will vary with geography and time, and it should be used in accordance with local official antibiotic prescribing guidelines and local susceptibility data

Indications include:

- Upper respiratory tract infections for example, ear, nose and throat infections, such as otitis media, sinusitis, tonsillitis and pharyngitis
- Lower respiratory tract infections for example, pneumonia , acute exacerbations of chronic obstructive pulmonary disease
- Genito-urinary tract infections for example, pyelonephritis, cystitis and urethritis
- Skin and soft tissue infections for example, furunculosis, pyoderma and impetigo
- Gonorrhoea, acute uncomplicated gonococcal urethritis and cervicitis

Pharmacology:

Pharmacodynamics

The prevalence of acquired resistance is geographically and time dependent and for select species may be very high. Local information on resistance is desirable, particularly when treating severe infections.

***In vitro* susceptibility of micro-organisms to Cefuroxime** Where clinical efficacy of cefuroxime axetil has been demonstrated in clinical trials this is indicated with an asterisk (*).
Commonly Susceptible Species
Gram-Positive Aerobes: *Staphylococcus aureus* (methicillin susceptible)* *Coagulase negative staphylococcus* (methicillin susceptible) *Streptococcus pyogenes** Beta-hemolytic streptococci
Gram-Negative Aerobes: *Haemophilus influenzae** including ampicillin resistant strains *Haemophilus parainfluenzae** *Moraxella catarrhalis** *Neisseria gonorrhoea** including penicillinase and non-penicillinase producing strains

Gram-Positive Anaerobes: *Peptostreptococcus* spp. *Propionibacterium* spp.
Spirochetes: *Borrelia burgdorferi**
Organisms for which acquired resistance may be a problem
Gram-Positive Aerobes: *Streptococcus pneumoniae**
Gram-Negative Aerobes: *Citrobacter* spp. not including *C. freundii* *Enterobacter* spp. not including *E. aerogenes* and *E. cloacae* *Escherichia coli** *Klebsiella* spp. including *Klebsiella pneumoniae** *Proteus mirabilis*
Proteus spp. not including *P. penneri* and *P. vulgaris* *Providencia spp.*
Gram-Positive Anaerobes: *Clostridium* spp.
Gram-Negative Anaerobes: *Bacteroides* spp. not including *B. fragilis* *Fusobacterium* spp.
Inherently resistant organisms
Gram-Positive Aerobes: *Enterococcus* spp. including *E. faecalis* and *E. faecium* *Listeria* monocytogenes
Gram-Negative Aerobes: *Acinetobacter* spp. *Burkholderia cepacia* *Campylobacter* spp. *Citrobacter freundii* *Enterobacter aerogenes* *Enterobacter cloacae* *Morganella morganii* *Proteus penneri* *Proteus vulgaris* *Pseudomonas* spp. including *P. aeruginosa* *Serratia* spp. *Stenotrophomonas maltophilia*
Gram-Positive Anaerobes: *Clostridioides difficile*
Gram-Negative Anaerobes: *Bacteroides fragilis*
Others: *Chlamydia* species *Mycoplasma* species *Legionella* species

Pharmacokinetics:

Absorption

After oral administration, *Cefuroxime axetil* is absorbed from the gastrointestinal tract and rapidly hydrolysed in the intestinal mucosa and blood to release cefuroxime into the circulation.

Absorption of cefuroxime is enhanced in the presence of food.

Following administration of *Cefuroxime axetil* tablets peak serum levels (2.1 mg/l for a 125 mg dose, 4.1 mg/l for a 250 mg dose, 7.0 mg/l for a 500 mg dose and 13.6 mg/l for a 1 g dose) occur approximately 2 to 3 hours after dosing when taken after food.

The rate of absorption of cefuroxime from the suspension compared with the tablets is reduced, leading to later, lower peak serum levels and reduced systemic bioavailability (4-17% less).

Distribution

Protein binding has been variously stated as 33-50% depending on the methodology used.

Metabolism

Cefuroxime is not metabolised.

Elimination

The serum half life is between 1 and 1.5 hours.

Cefuroxime is excreted by glomerular filtration and tubular secretion. Concurrent administration of probenecid increases the area under the mean serum concentrations time curve by 50%.

Recommended dose:

The dose of cefuroxime that is selected to treat an individual infection should take into account:

- The expected pathogens and their likely susceptibility to cefuroxime axetil
- The severity and the site of the infection
- The age, weight and renal function of the patient; as shown below.

The duration of therapy should be determined by the type of infection and the response of the patient, and should generally not be longer than recommended.

For optimal absorption *Cefuroxime axetil* should be taken after food.

- **Adults and Children weighing more than 40 kg**

Indication	**Dosage**
Acute tonsillitis and pharyngitis	250 mg twice daily
Acute otitis media	500 mg twice daily
Acute bacterial sinusitis	500 mg twice daily
Community acquired pneumonia	
Acute exacerbations of chronic obstructive pulmonary disease	
Urinary tract infections	250 mg – 500 mg twice daily
Uncomplicated gonorrhoea	Single dose of 1 g
Skin and soft tissue infections	250 mg – 500 mg twice daily

- **Children**

  There are no clinical trial data available on the use of *Cefuroxime axetil* in children under the age of 3 months.

Indication	Dosage
Acute tonsillitis and pharyngitis	10 mg/kg twice daily to a maximum of 500 mg daily
Acute otitis media	15 mg/kg twice daily to a maximum of 1000 mg daily
Acute bacterial sinusitis	
Community acquired pneumonia	
Urinary tract infections	
Skin and soft tissue infections	

The following two tables serve as a guideline for simplified administration from measuring spoons (5 ml) for the 125 mg/5 ml or the 250 mg/5 ml multi-dose suspension, and 125 mg single dose sachets.

10 mg/kg dosage

Weight range (kg)	Dose (mg) twice daily	No. of measuring spoons (5 ml) of sachets per dose	
		125 mg	250 mg
4 to 6	40 to 60	½	-
6 to 12	60 to 120	½ to 1	-
12 to 25	120 to 250	1 to 2	½ to 1
Greater than 25	250	2	1

15 mg/kg dosage

Weight range (kg)	Dose (mg) twice daily	No. of measuring spoons (5 ml) of sachets per dose	
		125 mg	250 mg
4 to 6	60 to 90	½	-
6 to 12	90 to 180	1 to 1½	½
12 to 16	180 to 240	1½ to 2	½ to 1
16 to 32	240 to 480	2 to 4	1 to 2
Greater than 32	500	4	2

To enhance compliance and improve the dosing accuracy in very young children, a dosing syringe can be supplied with a multidose bottle containing 50 ml of suspension. However, dosing in spoonfuls should be considered a more favourable option if the child is able to take the medication from the spoon.

If required, the dosing syringe may also be used in older children (please refer to the dosing tables below).

The recommended doses for the paediatric dosing syringe are expressed in ml or mg and according to bodyweight in the following tables:

10 mg/kg/dose (Paediatric dosing syringe)

Child's weight (kg)	Dose twice daily (mg)	125 mg/5 ml dose twice daily (ml)	250 mg/5 ml dose twice daily (ml)
4	40	1.6	0.8
6	60	2.4	1.2
8	80	3.2	1.6
10	100	4.0	2.0
12	120	4.8	2.4
14	140	5.6	2.8

15 mg/kg/dose (Paediatric dosing syringe)

Child's weight (kg)	Dose twice daily (mg)	125 mg/5 ml dose twice daily (ml)	250 mg/5 ml dose twice daily (ml)
4	60	2.4	1.2
6	90	3.6	1.8
8	120	4.8	2.4
10	150	6.0	3.0
12	180	7.2	3.6
14	210	8.4	4.2

- **Renal impairment**

Cefuroxime is primarily excreted by the kidneys. In patients with markedly impaired renal function it is recommended that the dosage of cefuroxime be reduced to compensate for its slower excretion (see the table below).

Creatinine Clearance	T ½ (hours)	Recommended Dosage
≥30 ml/min	1.4 – 2.4	No dose adjustment necessary standard dose of 125 mg to 500 mg given twice daily
10-29 ml/min	4.6	Standard individual dose given every 24 hours
<10 ml/min	16.8	Standard individual dose given every 48 hours
During haemodialysis	2 – 4	A single additional standard individual dose should be given at the end of each dialysis

Route of administration:Oral

Instructions for Use: Shake bottle to loosen granules. Add boiled and cooled water up to the mark on the bottle, Shake well, add additional boiled and cooled water, if necessary, to make up to the mark.

Contraindication: Hypersensitivity to cefuroxime or to any of the excipients. Patients with known hypersensitivity to cephalosporin antibiotics.
History of severe hypersensitivity (e.g. anaphylactic reaction) to any other type of beta lactam antibacterial agent (penicillins, monobactams and carbapenems).

Warnings and precaution:

Jarisch-Herxheimer reaction

The Jarisch-Herxheimer reaction has been seen following cefuroxime axetil treatment of Lyme disease. It results directly from the bactericidal activity of cefuroxime axetil on the causative bacteria of Lyme disease, the spirochaete *Borrelia burgdorferi.* Patients should be reassured that this is a common and usually self-limiting consequence of antibiotic treatment of Lyme disease. As with other antibiotics, use of cefuroxime axetil may result in the overgrowth of Candida. Prolonged use may also result in the overgrowth of other non-susceptible microorganisms (e.g. enterococci and *Clostridium difficile*), which may require interruption of treatment. Antibacterial agents–associated pseudomembranous colitis have been reported with nearly all antibacterial agents, including cefuroxime and may range in severity from mild to life threatening. This diagnosis should be considered in patients with diarrhoea during or subsequent to the administration of cefuroxime. Discontinuation of therapy with cefuroxime and the administration of specific treatment for *Clostridium difficile* should be considered. Medicinal products that inhibit peristalsis should not be given.

Interference with diagnostic tests

The development of a positive Coomb's Test associated with the use of cefuroxime may interfere with cross matching of blood. As false negative results may occur in the ferricyanide test, it is recommended that either the glucose oxidase or hexokinase methods are used to determine blood/plasma glucose levels in patients receiving cefuroxime axetil.

Beta-lactam antibiotics:

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with SANFUR DS – 125, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, SANFUR DS – 125 must be discontinued immediately and appropriate alternative therapy instituted.

Aspartame

Unsuitable for phenylketonurics.

Pregnancy and lactation:

Pregnancy: There is limited data from the use of cefuroxime in pregnant women. Studies in animals have shown no harmful effects on pregnancy, embryonal or foetal development, parturition or postnatal development. Cefuroxime for oral suspension should be prescribed to pregnant women only if the benefit outweighs the risk.

Breastfeeding: Cefuroxime is excreted in human milk in small quantities. Adverse effects at therapeutic doses are not expected, although a risk of diarrhoea and fungus infection of the mucous membranes cannot be excluded. Breastfeeding might have to be discontinued due to these effects. The possibility of sensitisation should be taken into account. Cefuroxime should only be used during breastfeeding after benefit/risk assessment by the physician in charge.

Effects on Ability to Drive and Use Machine: SANFUR – DS 125 can make you dizzy and have other side effects that make you less alert. Don't drive or use machines if you do not feel well.

Side effects:

Like all medicines, this medicine can cause side effects, although not everybody gets them. Conditions you need to look out for, A small number of people taking Cefuroxime Axetil get an allergic reaction or potentially serious skin reaction. Symptoms of these reactions include: severe allergic reaction. Signs include raised and itchy rash, swelling, sometimes of the face or mouth causing difficulty in breathing.

- Skin rash, which may blister, and looks like small targets (central dark spot surrounded by a paler area, with a dark ring around the edge).
- a widespread rash with blisters and peeling skin. (These may be signs of Stevens-Johnson syndrome or toxic epidermal necrolysis).

Other conditions you need to look out for while taking Cefuroxime Axetil include:

- fungal infections. Medicines like SANFUR – DS 125 can cause an overgrowth of yeast (Candida) in the body which can lead to fungal infections (such as thrush). This side effect is more likely if you take Cefuroxime Axetil for a long time.
- Severe diarrhoea (Pseudomembranous colitis). Medicines like SANFUR – DS 125 can cause inflammation of the colon (large intestine), causing severe diarrhoea, usually with blood and mucus, stomach pain, fever
- Jarisch-Herxheimer reaction. Some patients may get a high temperature (fever), chills, headache, muscle pain and skin rash while being treated with Cefuroxime Axetil for Lyme disease. This is known as the Jarisch-Herxheimer reaction. Symptoms usually last a few hours or up to one day.

Contact a doctor or nurse immediately if you get any of these symptoms. Common side effects

These may affect up to 1 in 10 people:

- fungal infections (such as Candida)
- headache
- dizziness
- diarrhoea
- feeling sick
- stomach pain.

Common side effects that may show up in blood tests:

- an increase in a type of white blood cell (eosinophilia)
- an increase in liver enzymes.

Uncommon side effects, these may affect up to 1 in 100 people:

- being sick
- skin rashes.

Uncommon side effects that may show up in blood tests:

- a decrease in the number of blood platelets (cells that help blood to clot)
- a decrease in the number of white blood cells
- Positive Coomb’s test.

Other side effects: Other side effects have occurred in a very small number of people, but their exact frequency is unknown:

- severe diarrhoea (pseudomembranous colitis)
- allergic reactions
- skin reactions (including severe)
- high temperature (fever)
- yellowing of the whites of the eyes or skin
- inflammation of the liver (hepatitis).

Side effects that may show up in blood tests:

- red blood cells destroyed too quickly (haemolytic anaemia)

Interactions with other medicines: Drugs which reduce gastric acidity may result in a lower bioavailability of cefuroxime axetil compared with that of the fasting state and tend to cancel the effect of enhanced absorption after food. Cefuroxime axetil may affect the gut flora, leading to lower oestrogen reabsorption and reduced efficacy of combined oral contraceptives.

Cefuroxime is excreted by glomerular filtration and tubular secretion. Concomitant use of probenecid is not recommended. Concurrent administration of probenecid significantly increases the peak concentration, area under the serum concentration time curve and elimination half-life of cefuroxime. Concomitant use with oral anticoagulants may give rise to increased INR (International Normalized Ratio).

Symptoms and treatment for overdose: Overdose can lead to neurological sequelae including encephalopathy, convulsions and coma. Symptoms of overdose can occur if the dose is not reduced appropriately in patients with renal impairment. Serum levels of cefuroxime can be reduced by haemodialysis and peritoneal dialysis.

Instruction for Use: Shake bottle to loosen granules. Add boiled and cooled water up to the mark on the bottle, Shake well, add additional boiled and cooled water, if necessary, to make up to the mark. Shake well before use.

Storage Condition: Keep container tightly closed. Store at a temperature below 30°C. Protect from light. The reconstituted suspension should be stored in a refrigerator at temperature 2-8°C for up to 7 days. Do not use SANFUR-DS 125 if it shows any sign of deterioration. Do not use this medicine after the expiry date.

Shelf Life: Unopened: 24 months stored below 30°C.
Reconstituted suspension: 7 days in a refrigerator at a temperature between 2 to 8°C.

Pack quantities: Available in translucent HDPE bottle of 50ml

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